

MOSTOFA HASIN MAHDI

hasinmahdi.hmr@gmail.com · +8801799407741 · Uttara, Dhaka, Bangladesh · [GitHub](#) · [LinkedIn](#) · [Portfolio](#)

SUMMARY

Final-year Computer Science and Engineering student with hands-on experience building backend APIs, data-driven applications, and AI/ML systems using Python, FastAPI, React, PostgreSQL, and Scikit-learn. Experienced in developing OCR pipelines, multi-tenant systems, API validation workflows, and machine-learning applications.

EDUCATION

Shanto-Mariam University of Creative Technology

B.Sc. in Computer Science and Engineering

CGPA: 3.96

Dhaka, Bangladesh

2022 – Expected Nov 2026

PROJECTS

NAB Preg AI

FastAPI, Next.js, XGBoost, Scikit-learn, Mistral AI, Supabase

Lead ML Engineer (Team of 5)

- Developed a single-pass ML model predicting 3 clinical risk targets in **under 50ms**, using WHO-aligned thresholds to address class imbalance.
- Developed an **OCR pipeline** integrating Mistral's **Pixtral-12B** via FastAPI, parsing prescriptions and lab reports into sanitized JSON for frontend validation.
- Engineered a **5-step OCR failover** pipeline using 4 backup Mistral API keys and Tesseract.js as a local fallback.
- Implemented dual-layer Zod/Pydantic validation to sanitize 10 patient fields and prevent malformed database payloads.
- Demo:* [Live Demo](#) · [GitHub](#)

PDF to Markdown Conversion Engine

PyMuPDF, pdfplumber, Pandas, spaCy, PyTesseract

Solo Developer

- Developed an open-source Python CLI using custom **coordinate-based** layout detection to sort text, detect headings, and prevent overlapping content.
- Integrated **spaCy-based text cleaning** to resolve sentence fragmentation and PyTesseract OCR for scanned pages.
- Engineered table extraction that identifies complex grids and reconstructs them into aligned Markdown tables.
- Demo:* [GitHub](#) · [Release](#)

Bizit Business Suite

React, FastAPI, PostgreSQL, RLS

Solo Developer

- Isolated data across **5 test tenants** in a single PostgreSQL schema using Row-Level Security (RLS), achieving **zero cross-tenant leaks**.
- Implemented 3-tier RBAC (Admin/Manager/Viewer) and Pydantic validation across 24 API endpoints.
- Designed finance dashboards aggregating sales, costs, and profit in **under 500ms**, replacing a 30-minute daily spreadsheet workflow.
- Demo:* username: hasin, password: hasin123 · [Live Demo](#) · [GitHub](#)

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL
Backend/Frontend: FastAPI, Next.js, React, REST APIs, Pydantic
AI/ML: Scikit-learn, XGBoost, Pandas, OCR, NLP
Databases: PostgreSQL, MongoDB, MySQL
DevOps & Tools: Docker, Git, GitHub Actions, Postman, Linux

ACHIEVEMENTS

- Champion**, SMUCT Software Project Showcase 2026; **Finalist**, CloudcampBD Infinity AI BuildFest 2026.
- 2nd Runner-up**, SMUCT CSE FEST V2.0 Programming Contest; participated in BUET & SUST IUPC Prelims and ICPC Regional Prelims (2022–2023).